

LESSON 6-8 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Using Properties to Rewrite Expressions**

Maryland Standard 6.EE.A.3 · Learning target – I can use the distributive property and like terms to rewrite an expression in simplest form.

REMEMBER

- **Distributive property:** multiply the outside number by **each** term inside: $4(x + 5) = 4x + 20$.
- **Commutative:** change the order. **Associative:** change the grouping.
- Simplest form: distribute, put like terms together, then combine them.

WORKED EXAMPLE

Write $3(k + 4) + 2k$ in simplest form.

Step 1 Distribute: $3k + 12 + 2k$

Step 2 Reorder (commutative): $3k + 2k + 12$

Step 3 Combine like terms: $5k + 12$

Answer $5k + 12$

WORD BANK

distribute – multiply the outside number by every term inside

like terms – terms with the same variable, like $4c$ and $5c$

simplest form – no like terms left to combine

PRACTICE

show your work · label units

1. Use the distributive property to rewrite each expression.

Step 1 Multiply the outside number by the first term.

Step 2 Multiply it by the second term too. Keep the + or – sign.

a. $6(n + 2) = \underline{\hspace{2cm}}$

b. $3(4x + 1) = \underline{\hspace{2cm}}$

c. $5(2 + y) = \underline{\hspace{2cm}}$

d. $7(a - 3) = \underline{\hspace{2cm}}$

e. $2(3b + 5) = \underline{\hspace{2cm}}$

f. $10(m - 1) = \underline{\hspace{2cm}}$

2. Use the distributive property to find each product. Show the sum you used.

a. $6 \times 34 = \underline{\hspace{2cm}}$

b. $8 \times 52 = \underline{\hspace{2cm}}$

3. Write each expression in simplest form.

a. $4(c + 3) + 5c = \underline{\hspace{2cm}}$

b. $2(x + 6) + x + 1 = \underline{\hspace{2cm}}$

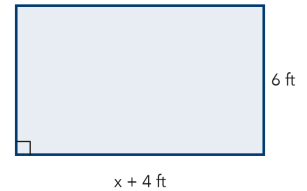
c. $3(m + 2) + 4(m + 1) = \underline{\hspace{2cm}}$

d. $8 + 5(t + 2) = \underline{\hspace{2cm}}$

4. **Garden.** A garden is 6 feet **wide** and $(x + 4)$ feet **long**. Its area is $6(x + 4)$ square feet. Write the area **without parentheses**. Then find the area when $x = 5$.

Step 1 Multiply 6 by x and 6 by 4.

Step 2 Put 5 in for x and find the value.



Show your work.

Answer _____

5. **Find the Error.** Ana simplified $2(3x + 4) + x$ and got $7x + 4$. Explain her mistake. Then simplify correctly.

Frame Ana forgot to multiply ____ by 2. The correct answer is ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Show someone at home how $3(10 + 2) = 30 + 6$ helps you find 3×12 in your head.

LESSON 6-8 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Expanding and Factoring Expressions**

Maryland Standard 6.EE.A.3 · Learning target – I can expand an expression and factor one using the greatest common factor.

REMEMBER

- **Expand:** distribute to remove parentheses: $5(2x + 3) = 10x + 15$.
- **Factor:** put the GCF outside the parentheses; divide each term by it.
- Check a factored form by expanding it.

WORKED EXAMPLEFactor $24y + 36$.

Step 1 The GCF of 24 and 36 is 12. Divide: $24y \div 12 = 2y$ and $36 \div 12 = 3$

Step 2 Write $12(2y + 3)$. Check: $12 \times 2y + 12 \times 3 = 24y + 36$

Answer $12(2y + 3)$ **WORD BANK****expand** – multiply to remove the parentheses**factor** – write as a number times (____ + ____)**GCF** – the biggest number that divides both numbers**PRACTICE***show your work · label units***1.** Expand each expression.

a. $3(5x + 2) = \underline{\hspace{2cm}}$

b. $8(y + 7) = \underline{\hspace{2cm}}$

c. $4(3 + 2n) = \underline{\hspace{2cm}}$

2. Factor each expression using the GCF.**Step 1** Find the GCF of the two numbers.**Step 2** Divide each term by the GCF.**Step 3** Write GCF(____ + ____) and check by expanding.

a. $6x + 15 = \underline{\hspace{2cm}}$

b. $20y + 32 = \underline{\hspace{2cm}}$

c. $14a + 49 = \underline{\hspace{2cm}}$

d. $36 + 27k = \underline{\hspace{2cm}}$

e. $40m + 60 = \underline{\hspace{2cm}}$

f. $11b + 33 = \underline{\hspace{2cm}}$

3. Rectangle. A rectangle has an **area** of $8x + 56$ square units. Its **width** is 8 units. Write an expression for its **length**.

*Show your work.***Length** _____

4. **Stretch.** Write two different factored forms of $12x + 24$. Then write the form that uses the GCF.

5. **Reason.** Leon factored $30n + 45$ as $5(6n + 9)$. Did he use the GCF?

Frame $6n$ and 9 share the factor _____. So Leon _____ use the GCF. The GCF form is _____.

☐ Yes

☐ No

TEST PRACTICE

choose the best answer

6. Which is $16x + 40$ factored using the GCF?

Step 1 Find the GCF of 16 and 40.

Step 2 Pick the choice with the GCF outside the parentheses.

A $2(8x + 20)$

B $8(2x + 5)$

C $4(4x + 10)$

D $8(2x + 40)$

7. Which expression is equivalent to $6(3a + 2)$?

A $18a + 2$

B $9a + 8$

C $18a + 12$

D $3a + 12$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Show someone at home how to factor $6x + 9$ as $3(2x + 3)$. Then expand it to check.